

Session 2 Quiz - AI Concepts & Terminology

Track B Grades 9–12 20 Questions

Q1 [Type: MCQ Difficulty: Easy | QID : 52447]

_____ is the process where an AI system learns from data - like studying flashcards over and over until you can recognise a pattern reliably. It involves making a guess, checking it, adjusting, and repeating millions of times.

- A. Sensing
- B. Training
- C. Predicting
- D. Filtering

Correct Answer: A

Explanation: Training is the process by which an AI system learns from data. The algorithm goes through a continuous loop - make a guess, check whether the guess was right, adjust its internal parameters by a tiny amount, and repeat - sometimes millions of times until accuracy is high.

Q2 [Type: MCQ Difficulty: Easy | QID : 52451]

An _____ is a step-by-step set of instructions to solve a problem or complete a task - like a recipe written for a computer.

- A. Output
- B. Engine
- C. Algorithm
- D. Interface

Correct Answer: C

Explanation: An algorithm is a finite sequence of well-defined instructions. There are four properties of any algorithm: clear (each step is precisely defined), unambiguous (no confusion about what to do), finite (it eventually ends), and effective (it actually solves the intended problem). Your morning routine, directions to a friend's house, and a baking recipe are all everyday algorithms.

Q3 [Type: MCQ Difficulty: Medium | QID : 52456]

After an AI algorithm completes the training process - learning patterns from a dataset - the resulting system, which is ready to make predictions on new, unseen information, is called a _____.

- A. Dataset
- B. Model
- C. Algorithm
- D. Feature

Correct Answer: B

Explanation: A model is the output of the training process - the algorithm applied to data until patterns are learned. Think of it as the skilled chef who has practiced a recipe so many times they can now cook it from memory. The dataset is the collection of training examples. The algorithm is the learning method. A feature is a specific measurable characteristic in the data. A model is the end product: a trained system ready to make predictions.

Q4 [Type: MCQ Difficulty: Easy | QID : 52460]

Machine Learning (ML), Deep Learning (DL), and Generative AI (GenAI) are all subsets of the broader field of Artificial Intelligence (AI).

A. True

B. False

Correct Answer: A (True)

Explanation: AI is the big bin containing everything related to making machines smart; ML is a smaller box inside AI that holds learning-from-data approaches; DL is an even smaller case inside ML that uses brain-like neural networks; and GenAI is a special compartment that holds the AI that creates new things. Your Google Storybook is simultaneously all four levels - it is AI, ML, DL, and GenAI at the same time.

Q5 [Type: MCQ Difficulty: Medium | QID : 52461]

When an AI makes a prediction, it is genuinely understanding the situation the same way a human would.

A. True

B. False

Correct Answer: B (False)

Explanation: Prediction is not understanding. AI finds patterns and calculates the most likely output based on its training data - it does not "know" what the words or images mean. When Gemini completes "The cat sat on the..." with "mat," it is not picturing a cat or a mat; it is calculating which word most often follows that phrase in its training data. This is also why AI can be confidently wrong, and why a human must always stay responsible for checking the output.

Q6 [Type: MCQ Difficulty: Medium | QID : 52463]

If you give two teams the exact same algorithm, they are guaranteed to end up with the exact same AI model, regardless of what data they used to train it.

A. True

B. False

Correct Answer: B (False)

Explanation: Same algorithm + different data = different models. This is also why "the data pipeline is the product, not the model": what data was collected, how it was labelled, what was missing, who was included or excluded - these choices shape everything about how the final model behaves.

Q7 [Type: MCQ Difficulty: Easy | QID : 52466]

Which subset of AI is specifically responsible for creating "new" content such as images, music, or code?

A. Deep Learning

B. Narrow AI

C. General AI

D. Generative AI

Correct Answer: D

Explanation: The prefix "Generative" refers to the ability to generate or create something new from scratch. GenAI is a subset of Deep Learning, which is itself a subset of Machine Learning, which is a subset of AI.

Q8 [Type: MCQ Difficulty: Easy | QID : 52468]

Which of the following is an example of "Sensor Data"?

- A.** The GPS location and movement speed of a car
- B.** A photograph of a family at the beach
- C.** A transcript of a podcast episode about dinosaurs
- D.** A high score recorded in a video game

Correct Answer: A

Explanation: Sensor data includes readings from physical devices like GPS, motion trackers, and thermometers. It powers Google Maps traffic prediction, smartwatch fitness tracking, and self-driving cars. The other options describe image data, audio data, and numerical data - also valid data types, but not "sensor data" specifically.

Q9 [Type: MCQ Difficulty: Medium | QID : 52472]

What does it mean for an algorithm to be "finite"?

- A.** It has no confusing steps or ambiguous instructions
- B.** It can be run on any computer regardless of its age
- C.** It is written in a language that only humans can understand
- D.** It eventually reaches an end after a certain number of steps

Correct Answer: D

Explanation: An algorithm must solve a problem in a limited amount of time and steps, rather than running forever. "Finite" is one of the four properties of a valid algorithm, alongside clear, unambiguous, and effective.

Q10 [Type: MCQ Difficulty: Medium | QID : 52474]

Which of these is NOT one of the characteristics of a valid algorithm?

- A. Effective
- B. Creative
- C. Unambiguous
- D. Clear

Correct Answer: B

Explanation: Algorithms are logical and procedural; they follow set rules rather than possessing human-like creativity. The four required properties are clear, unambiguous, finite, and effective.

Q11 [Type: MCQ Difficulty: Medium | QID : 52477]

A recipe tells a cook exactly which ingredients to use and which steps to follow. A skilled chef who has practiced that recipe hundreds of times can now execute it perfectly from memory and adapt it confidently when an ingredient is unavailable. Which AI concepts do the recipe and the skilled chef best represent, respectively?

- A. Training Data and Prediction
- B. Dataset and Algorithm
- C. Algorithm and Model
- D. Model and Algorithm

Correct Answer: C

Explanation: The recipe is the algorithm - a precise set of instructions defining the method. The skilled chef is the model - the result of applying the algorithm through extensive practice on data until expertise is internalized. The chef has learned the recipe so thoroughly that they can apply it flexibly to new situations. This is exactly how a trained model works: it applies learned patterns to new inputs it has never seen before.

Q12 [Type: MCQ Difficulty: Medium | QID : 52478]

What happens inside an AI during the training loop, repeated millions of times?

- A. The AI deletes its data and starts fresh each round
- B. The AI memorises every example word-for-word so it never forgets anything
- C. The AI makes a guess, checks it, adjusts slightly, and repeats
- D. The AI asks a human to label every example by hand

Correct Answer: C

Explanation: Training is a continuous loop: guess → check → adjust → repeat. The "adjust" step changes internal parameters - internal numbers inside the algorithm - by very small amounts each time.

Q13 [Type: MCQ Difficulty: Medium | QID : 52483]

When Gmail predicts the next word as you type an email, or Netflix predicts which show you might watch next, what is the AI actually doing?

- A. Calculating the most likely output from learned patterns
- B. Thinking the way a human reader thinks
- C. Asking another AI for the answer in real time
- D. Reading your mind through the keyboard

Correct Answer: A

Explanation: Every prediction is a statistical guess based on patterns from training data. Gmail Smart Compose predicts the most likely next word given what you have typed so far. Netflix predicts what you will watch based on patterns from millions of similar viewers. This is also why AI can be confidently wrong - a confident prediction is still just a prediction, not a fact, which is why humans must remain responsible for verifying important information.

Q14 [Type: MCQ Difficulty: Medium | QID : 52485]

The AI Recipe" lays out the components of any AI system in a specific order. Which of the following correctly reflects that order?

- A. Prediction → Model → Training → Algorithm → Data
- B. Data + Algorithm → Training → Model → Prediction
- C. Model → Data → Algorithm → Prediction → Training
- D. Training → Data → Prediction → Algorithm → Model

Correct Answer: B

Explanation: The AI Recipe is: Data + Algorithm → Training → Model → Prediction. Data and an algorithm are the starting materials. Training is the process that turns them into a trained model. The model is then used to make predictions on new, unseen data. This sequence applies to every AI system - YouTube recommendations, ChatGPT, self-driving cars, spam filters - even though the details differ enormously.

Q15 [Type: MCQ Difficulty: Hard | QID : 52486]

Which of the following BEST describes the difference between an algorithm and a model?

- A. They are the same thing, just different words
- B. The model exists before the algorithm
- C. An algorithm only works on small data, while a model works on big data
- D. An algorithm is the general method; a model is the result after training on data

Correct Answer: D

Explanation: The algorithm is general and reusable - it can be applied to many problems. The model is specific - it is what an algorithm becomes after training on a particular dataset. The same algorithm trained on different data produces different models, which is why two companies can use the same learning method and end up with very different AI products. The algorithm alone is just instructions; the model is the result.

Q16 [Type: MCQ Difficulty: Hard | QID : 52488]

If an AI system trained on historical data ends up producing biased outputs, where does the bias most often come from?

- A.** From the historical data, which carries patterns from past human decisions
- B.** From bugs in the algorithm's code that introduce errors during training
- C.** From the model gradually drifting in new directions after deployment
- D.** From the hardware that runs the model overheating and corrupting outputs

Correct Answer: A

Explanation: Bias in AI almost never comes from the algorithm itself - algorithms are general methods with no preferences. It comes from the data the algorithm was trained on. If historical hiring records, loan decisions, or medical referrals contain past human bias, the model learns those patterns as if they were reliable signals. This is why the session says "the data pipeline is the product, not the model" and why GIGO matters so much. The fix is never to rewrite the algorithm or replace the hardware - it is to fix the data pipeline by correcting labels, sampling more representatively, or rethinking what outcome the model is being trained to predict.

Q17 [Type: MCQ Scenario Difficulty: Medium | QID : 52491]

Scenario: When you set up Face Unlock on your phone, the phone takes thousands of photos of your face from different angles and in different lighting. Later, your phone can recognise you even when you wear glasses, a hat, or a new haircut.

Question: In the "AI Recipe," what is the personalised system that recognises your face called?

- A.** The algorithm
- B.** The training set
- C.** The trained model
- D.** The hardware

Correct Answer: C

Explanation: The photos of your face are the data. The face-recognition algorithm is the general method for learning to identify faces. The training process is what happens when the algorithm goes through your face photos again and again, adjusting its parameters. The trained model is the final, personalised result - the system that can now recognise YOUR face specifically. The same algorithm would produce a different model on a different person's face data.

Q18 [Type: MCQ Scenario Difficulty: Hard | QID : 52494]

Scenario: A university deploys an AI system to predict which first-year students are at risk of dropping out before completing their degree. The model was trained on five years of student records including grades, attendance, financial aid status, and extracurricular activity. The system flags "high risk" students for counseling outreach. An administrator says: "The model makes accurate predictions - so it must understand why students drop out."

Question: What is wrong with the administrator's claim?

- A.** The administrator is correct - a model trained on five years of records has enough historical data to understand the causes of student dropout
- B.** The administrator is wrong - predictive models cannot make predictions about individual students, only about aggregate groups
- C.** The administrator is correct - making consistently accurate predictions about an outcome is functionally equivalent to understanding why that outcome occurs
- D.** The administrator is wrong - the model identifies statistical correlations between measurable factors and dropout outcomes but has no understanding of the reasons

Correct Answer: D

Explanation: Prediction and understanding are fundamentally different cognitive operations. The model has found that certain measurable patterns statistically precede dropout. It has no concept of causation, no awareness of individual context, and no ability to reason about what would change the trajectory for a specific student. Accurate statistical prediction is not the same as causal comprehension.

Q19 [Type: FRQ - Short Answer Difficulty: Hard | QID :]

A team that built an AI model to evaluate small business loan applications reports: "Our model achieves 89% accuracy on our test set it is ready for deployment."

Identify one specific question about the test data that must be answered before this claim of readiness can be trusted, and explain precisely why the answer to that question determines whether the 89% figure is meaningful.

(Respond in 3–5 sentences.)

Model Answer: The team must ask: does the test set represent the same diversity of applicants the model will encounter in real deployment including businesses from all geographic regions, industries, ownership demographics, and size ranges that will actually apply for loans? An 89% accuracy figure only tells you how the model performs on data drawn from the same distribution as its training data if the test set was sampled from the same limited historical pool, the accuracy reflects performance on familiar patterns, not real-world generalization. A model that achieves 89% accuracy on businesses from its training context but performs at 60% on businesses from underrepresented industries or demographics is not deployment-ready and the 89% figure would never reveal that gap if the test set does not include those groups. Accuracy is a conditional measurement: it describes performance under the specific conditions used to evaluate it, and those conditions must match the deployment context for the number to be meaningful.

Q20 [Type: FRQ - Scenario Analysis Difficulty: Hard | QID :]

An online learning platform trains a model to predict which enrolled students are likely to drop out before completing their course. The platform uses this prediction to automatically reduce email communication with students flagged as "likely to drop out," explaining that messaging these students wastes marketing resources since they will probably leave anyway.

Identify the most significant problem with this deployment decision. Then describe what a better use of the exact same model and predictions would look like.

(Respond in 4–6 sentences.)

Model Answer: The most significant problem is that the platform is using a prediction of struggle as a reason to withdraw support when the same prediction should trigger increased support. A student flagged as "likely to drop out" is a student who is struggling, and reducing contact with that student makes dropout more likely, not less creating a self-fulfilling prophecy where the model's prediction is confirmed partly because of the platform's response to it. This inverts the purpose of having predictive information: the model is being used to optimize marketing spend rather than to improve student outcomes, treating the students with the highest need as the least worth serving. The same model and the same predictions, used differently, would look like this: a "likely to drop out" flag triggers immediate, personalized outreach a check-in message from an instructor, an offer of additional academic support resources, or a prompt connecting the student to a peer community. The prediction becomes the beginning of an intervention rather than the justification for abandoning one. The value of predictive information is not in confirming who will fail it is in identifying who can be helped before failure occurs.